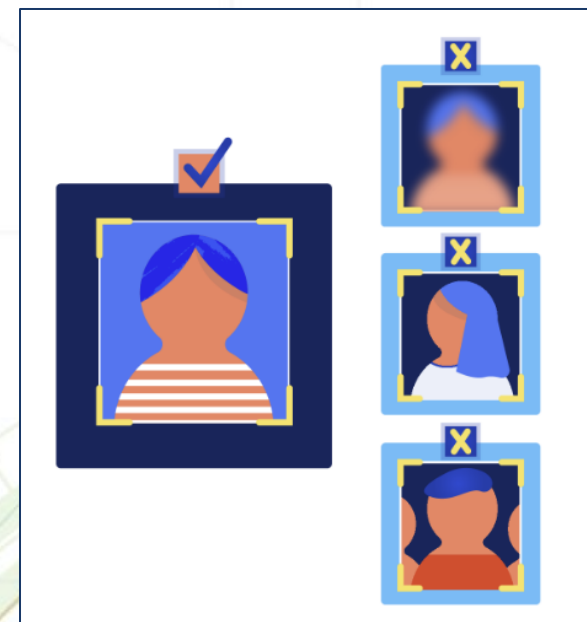


NIST FATE Quality Specific Image Defect Detection (SIDDD)

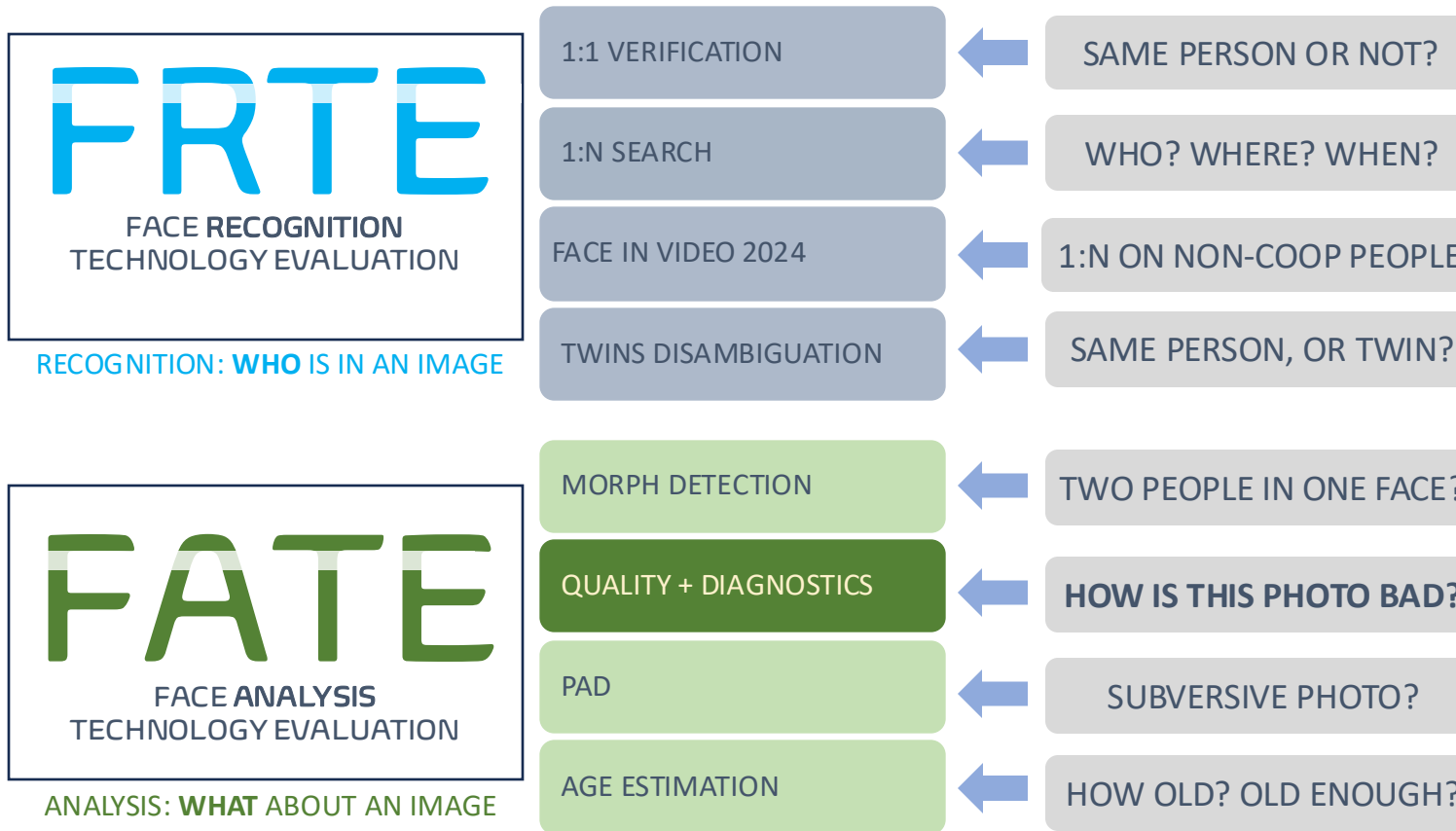
Joyce Yang-NIST Image Group

IFPC 2025: Virtual Presentation, April 1 2025

FRTE/FATE Team: Patrick Grother, Mei Ngan, Kayee Hanaoka, Austin Hom, Joyce Yang, Jim Matey



NIST Face Benchmarks



FRTE/FATE evaluations are:

fast transparent fair
IP-protecting robust open regular
large-scale free independent
extensible public
repeatable global closed-box

Why Specific Image Defect Detection?

- Automated face recognition is sensitive to quality problems in images
- Standards set requirements on images
 - For MRTD, ISO/IEC:39794-5
 - For law enforcement, ANSI/NIST standard
- SIDD gives users a set of tools to check those requirements

Shadows and Lighting

Glasses

Size and Position

Resolution, Print Size, and Quality

Digital Alterations and Retouching

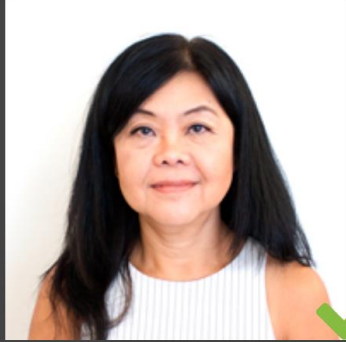
Pose and Expression

Attire, Hats, and Hair

Background

Children

Online Renewal Application

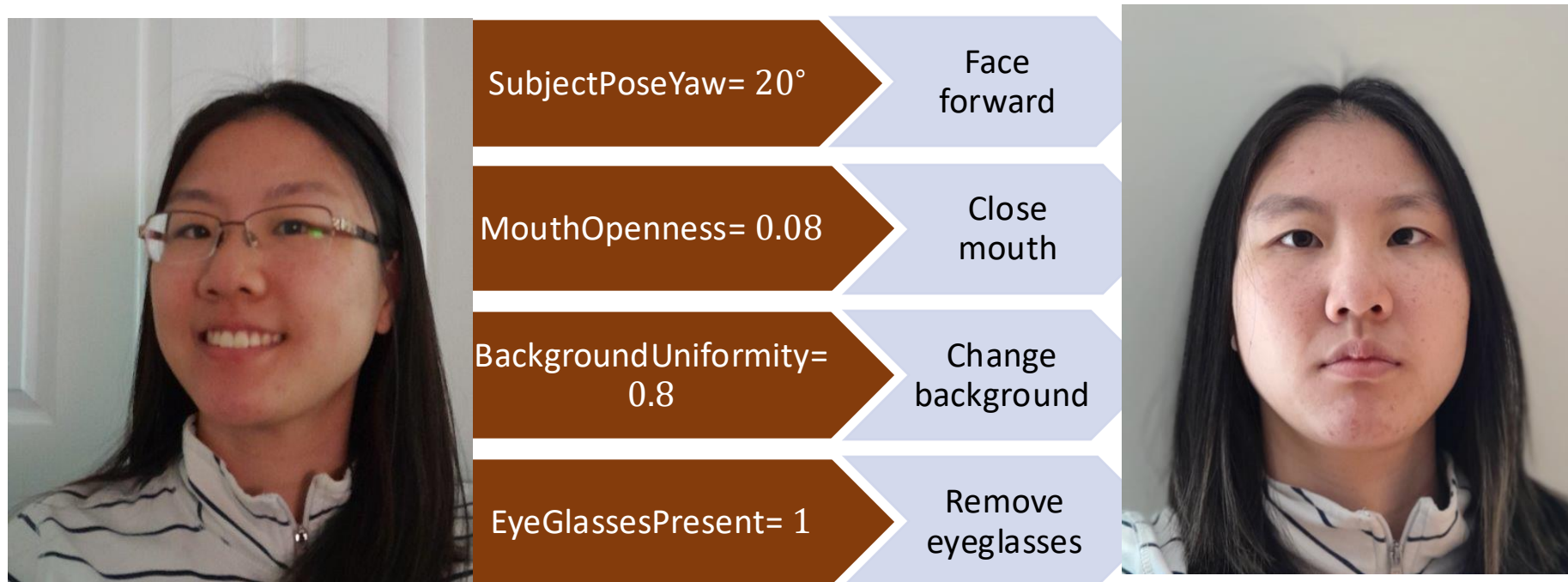


Acceptable— Background is white or off-white, without shadows, and is plain without texture, objects, or lines

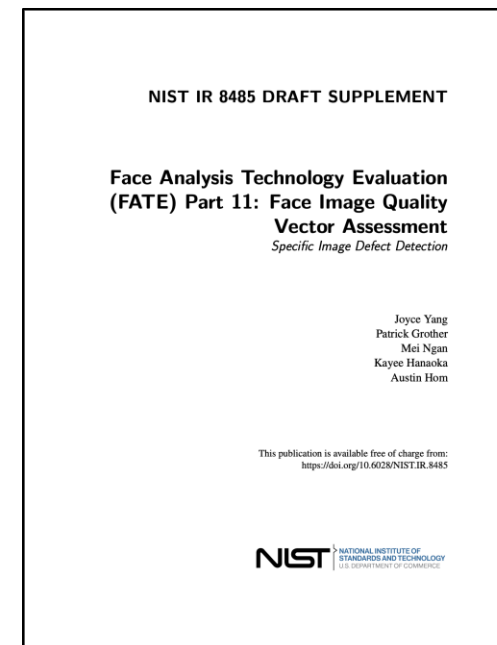


Why SIDD? Example: Passport Photo

Face capture software should be able to point out an issue such as the mouth being open and give feedback in real time. A SIDD algorithm might provide the output below.



FATE SIDD At A Glance



The two main goals of SIDD are to

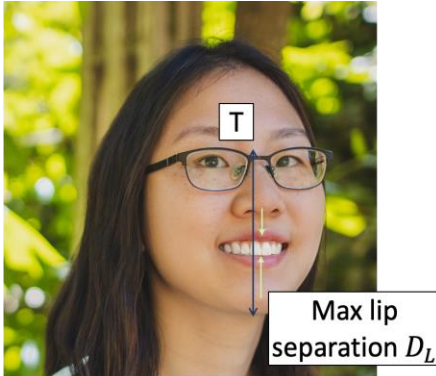
- 1) Promote quality defect detection algorithm development and
- 2) Support ISO/IEC 29794-5 on face image quality.

Participation By Year

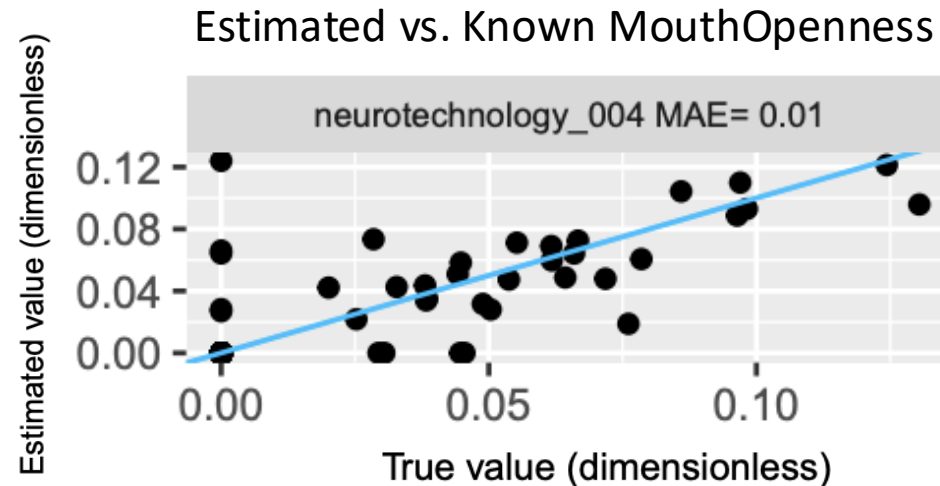
Year	Number of Algorithms	Unique Developers
2022	2	2
2023	15	9
2024	23	9
2025	1	1

Subject-Related Quality Measure Example: Mouth Openness

$$D_{mouthOpen} = \frac{D_L}{T}$$



The Mouth Openness measure, MouthOpen2, is defined as the (vertical) maximum lip separation normalized by the chin-to-eyes distance.



Mean Absolute Error (MAE) is calculated over all images.

The blue line $y=x$ represents perfect performance.

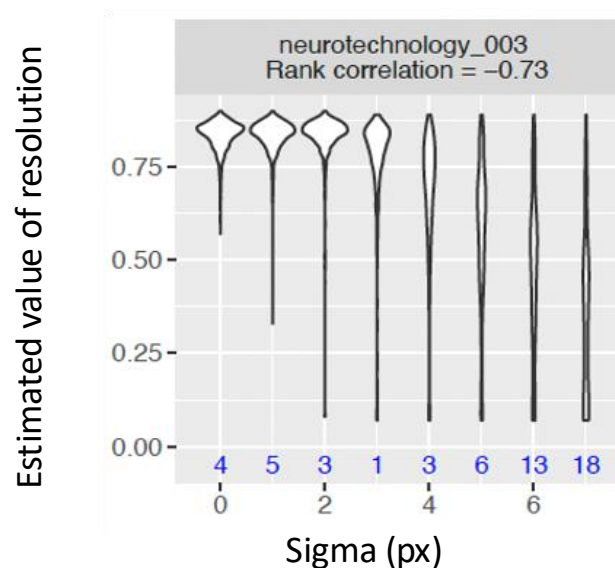
Algorithm	MAE (dimensionless)
neurotechnology_005	0.00
neurotechnology_004	0.01
mobbl_002	0.01
igd_003	0.01
secunet_003	0.01

Neurotechnology_005 performs best based on mean absolute error.

Capture-Related Quality Measure Example: Resolution

Standard deviation σ	0	4	7
Result of convert – gaussian-blur $0 \times \sigma$			

The resolution sets are made by applying synthetic blur. Higher values of sigma correspond to lower resolution.

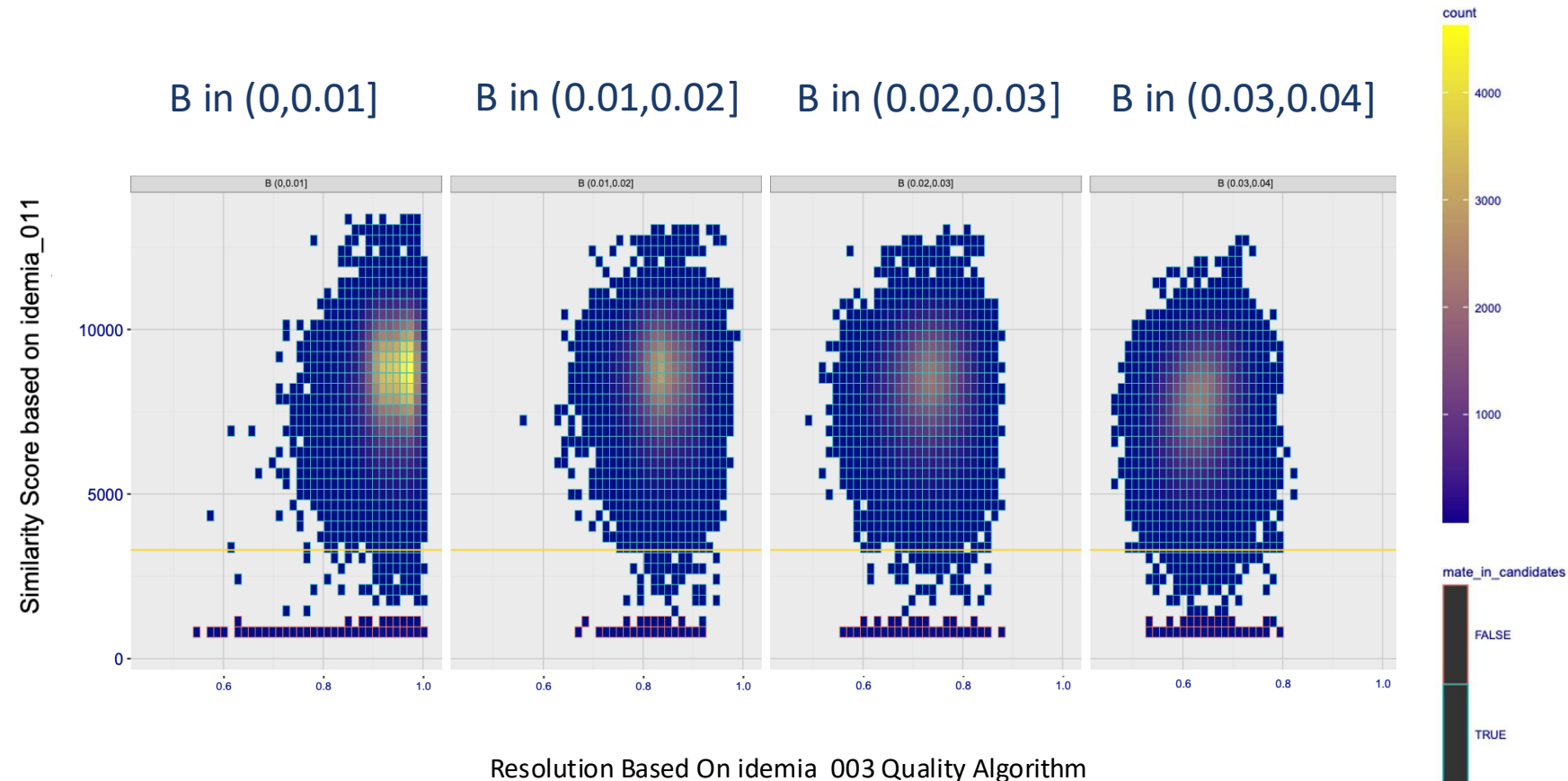


We calculate rank correlation over all images. Perfect performance corresponds to a value of -1.

Preliminary Study of Resolution Effect on Recognition

➤ Each panel corresponds to a range of $B = \text{sigma} / \text{IED}$ i.e. the amount of blur “damage.”

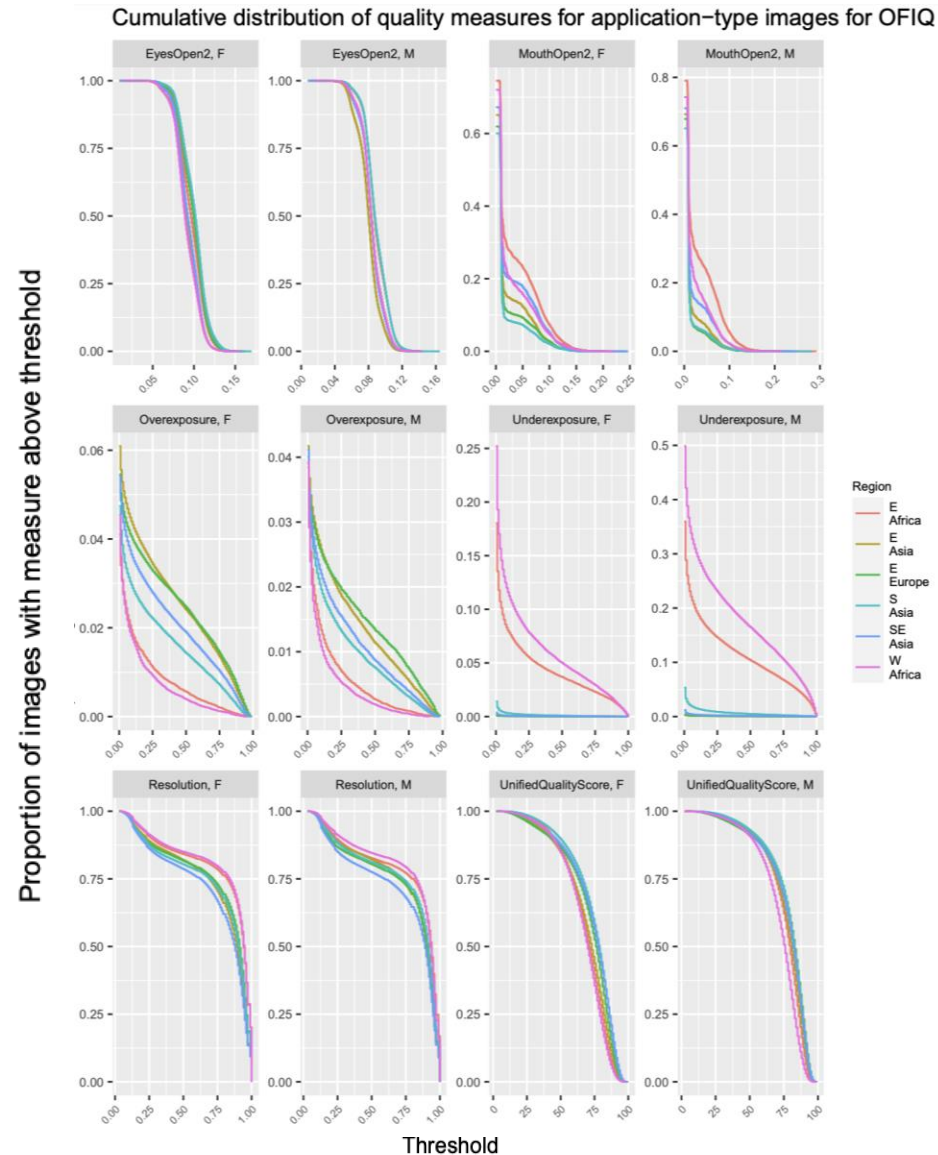
➤ The mated similarity score cloud moves downward and leftward as B increases.



OFIQ Evaluation

OFIQ is the reference implementation of ISO/IEC 29794-5.

We contributed analysis of OFIQ to the WG3 Ad Hoc Group Report.

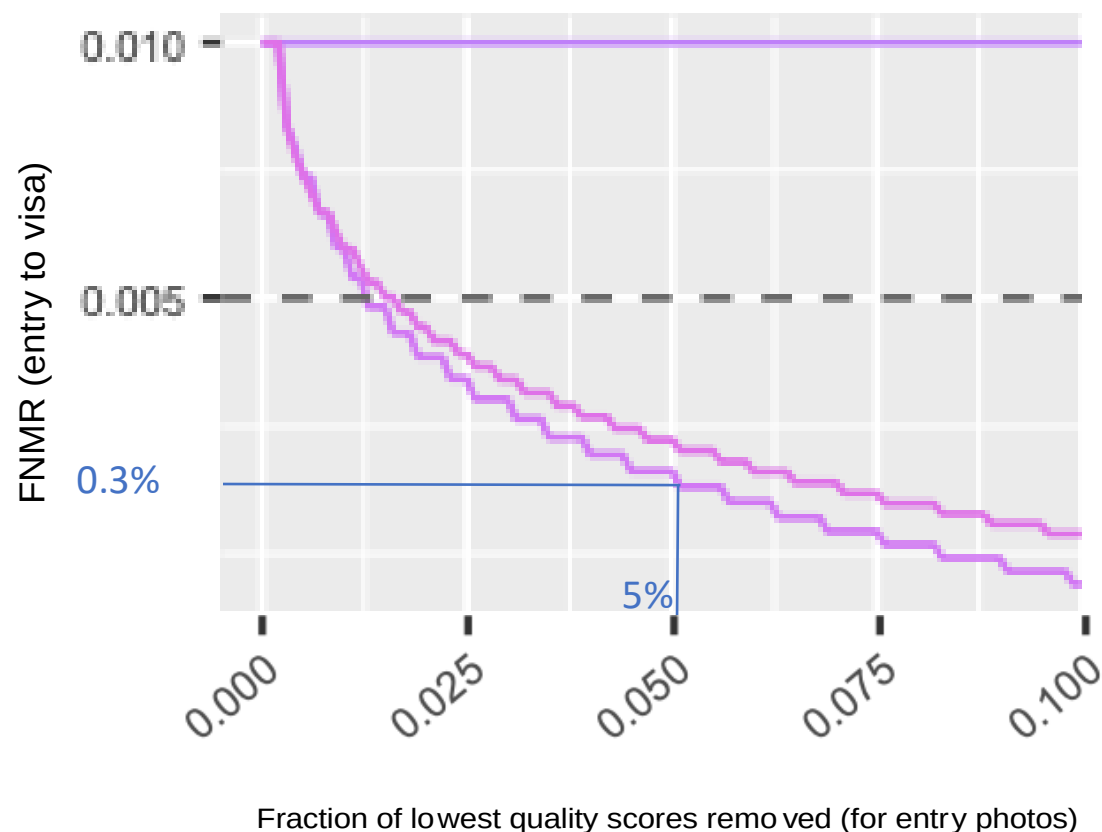


Measure	Component Algorithm
EyesOpen2	Secunet_003
MouthOpen2	Secunet_003
Overexposure	Secunet_005
Underexposure	Secunet_005
Resolution	Secunet_003
Unified Quality Score	Secunet_003

Evaluation of Unified Quality Score

- For Unified Quality Score (UQS), quality prediction of false non-match rates is shown by the Error Discard Characteristic (EDC).
- This has been used for face quality in our 2019 quality evaluation and for fingerprint prior to that.
- False non-match rate (FNMR) is calculated based on consensus of 15 recognition algorithms. Lower FNMR is better.
- EDC curves are shown for secunet_002, secunet_004, secunet_003 (from top to bottom)
- Secunet_003 UQS is used in OFIQ.

Quality as a Predictor of Genuine Scores: Secunet

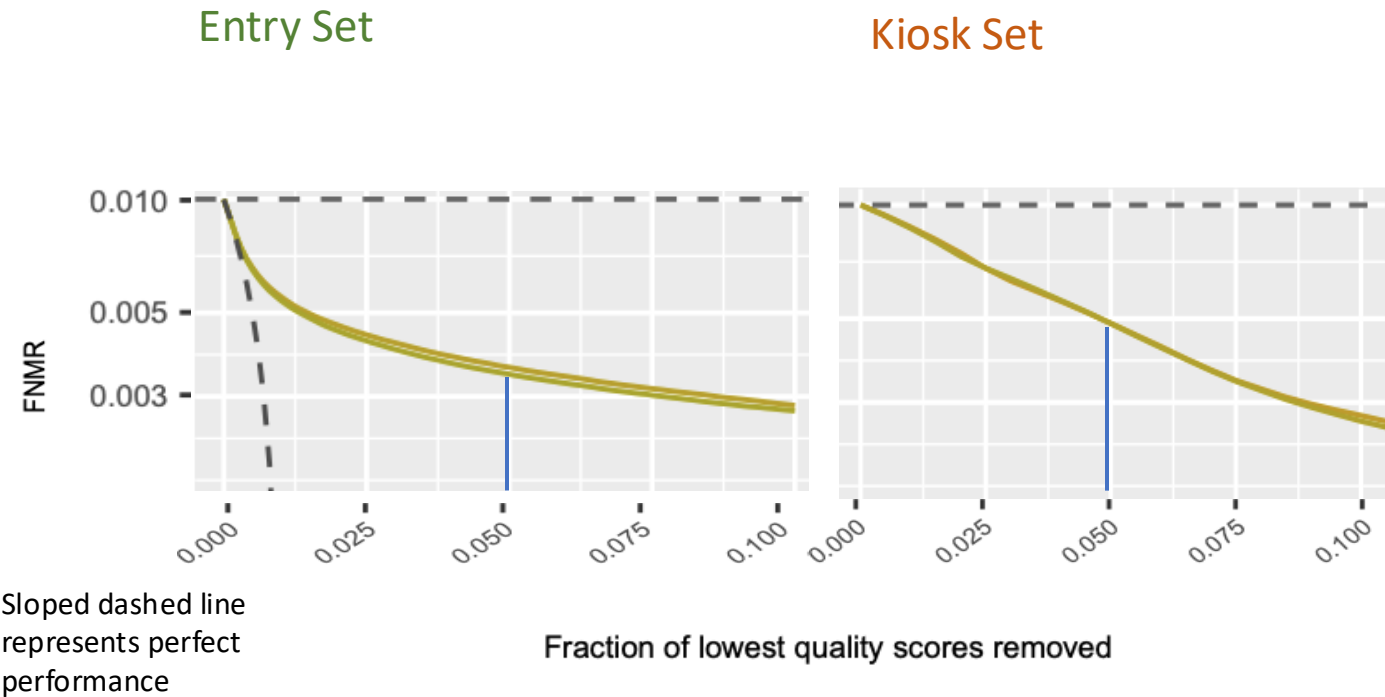


Kiosk Set For Unified Quality Score: Error Curves

Algorithms perform worse for the **Kiosk** set than for the **Entry** set.

Kiosk generally yields higher FNMR than Entry for low discard fractions.

Images have varying pose, illumination, and background clutter.



EDC curves for **Entry** (left; higher quality) and **Kiosk** (right; lower quality) for idemia_002,_003.

FNMR is calculated based on consensus of 15 recognition algorithms.

Kiosk Set for Unified Quality Score: Leaderboard



Kiosk set generally yields higher FNMR than the Entry set for discard fraction of 5%.

Entry Set

Quality Algorithm	FNMR after removal of 5% lowest quality Entry images. Initial FNMR = 0.01. Lower is better.
secunet_003	0.0031 ⁽¹⁾
igd_005	0.0032 ⁽²⁾
qazsmartvisionai_000	0.0033 ⁽³⁾
secunet_004	0.0034 ⁽⁴⁾
intema_000	0.0034 ⁽⁵⁾
idemia_003	0.0034 ⁽⁶⁾
idemia_001	0.0034 ⁽⁷⁾
frpkauai_001	0.0034 ⁽⁸⁾
idemia_002	0.0036 ⁽⁹⁾
datech_000	0.0036 ⁽¹⁰⁾

Kiosk Set

Quality Algorithm	FNMR after removal of 5% lowest quality Kiosk images. Initial FNMR = 0.01. Lower is better.
idemia_003	0.0049 ⁽¹⁾
idemia_002	0.0049 ⁽²⁾
datech_000	0.0051 ⁽³⁾
neurotechnology_003	0.0057 ⁽⁴⁾
roc_008	0.0062 ⁽⁵⁾
papil11_000	0.0062 ⁽⁶⁾
igd_005	0.0063 ⁽⁷⁾
neurotechnology_005	0.0064 ⁽⁸⁾
neurotechnology_004	0.0064 ⁽⁹⁾
roc_007	0.0065 ⁽¹⁰⁾

- We are in the process of studying effect of SIDD resolution on recognition
- We evaluated OFIQ, the reference implementation for ISO/IEC 29794-5.
- We are in the process of adding a challenging set for Unified Quality Score
- FATE SIDD remains open; 4-month frequency for participation

- For more information and updates, see https://pages.nist.gov/frvt/html/frvt_quality.html or scan the QR code.



Questions?

Contact frvt@nist.gov

Visit https://pages.nist.gov/frvt/html/frvt_quality.html

View SIDD report at

https://pages.nist.gov/frvt/reports/quality_sidd/frvt_quality_sidd_report.pdf

Additional Information

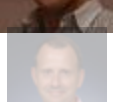
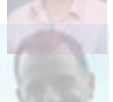
Complete Mapping of OFIQ to Secunet algorithms



Measure	Component Algorithm
EyesOpen2	Secunet_003
MouthOpen2	Secunet_003
Overexposure	Secunet_005
Underexposure	Secunet_005
Resolution	Secunet_003
Unified Quality Score	Secunet_003
BackgroundUniformity	Secunet_003
Compression Artifacts	Secunet_004
Inter-Eye Distance	Secunet_004
Face occlusion	Secunet_003
Head Pose	Secunet_004

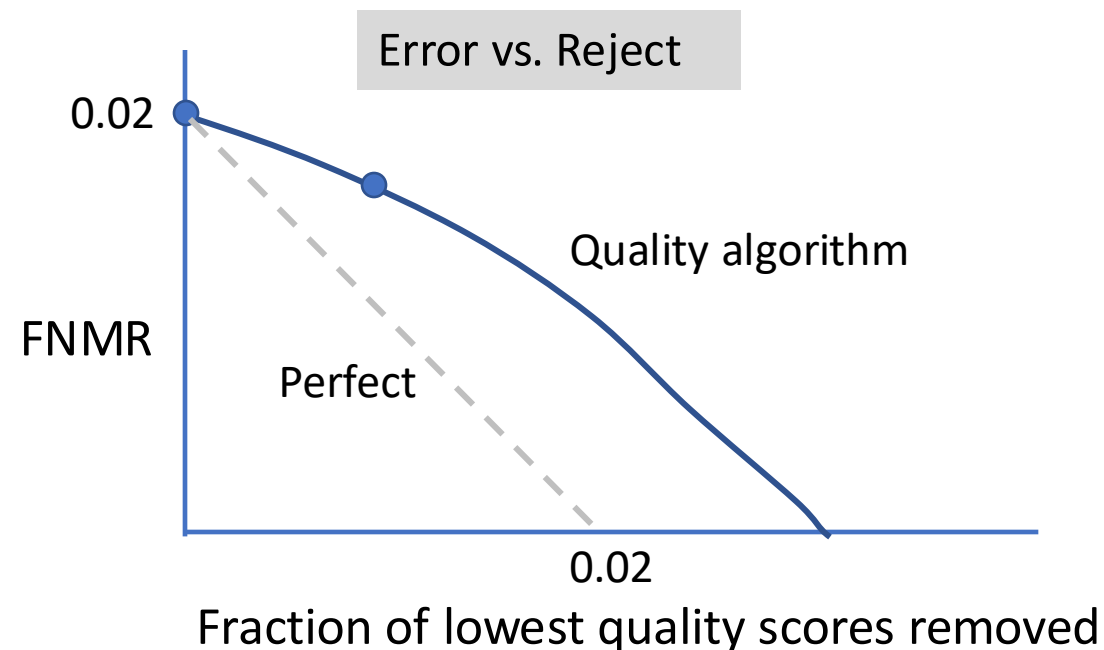
Quality Values as Predictors of Genuine Scores– Error vs. Reject

Quality algorithm

Ref	Probe	Quality
		97
		94
		74
		68
		57
		32
		29
		27

Recognition algorithm

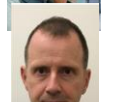
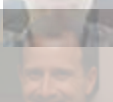
MATCH SCORE	MATCH YES?
0.97	TRUE
0.91	TRUE
0.60	FALSE
0.85	TRUE
0.81	TRUE
0.72	FALSE
0.90	TRUE
0.57	FALSE



Discard n% of lowest quality probes

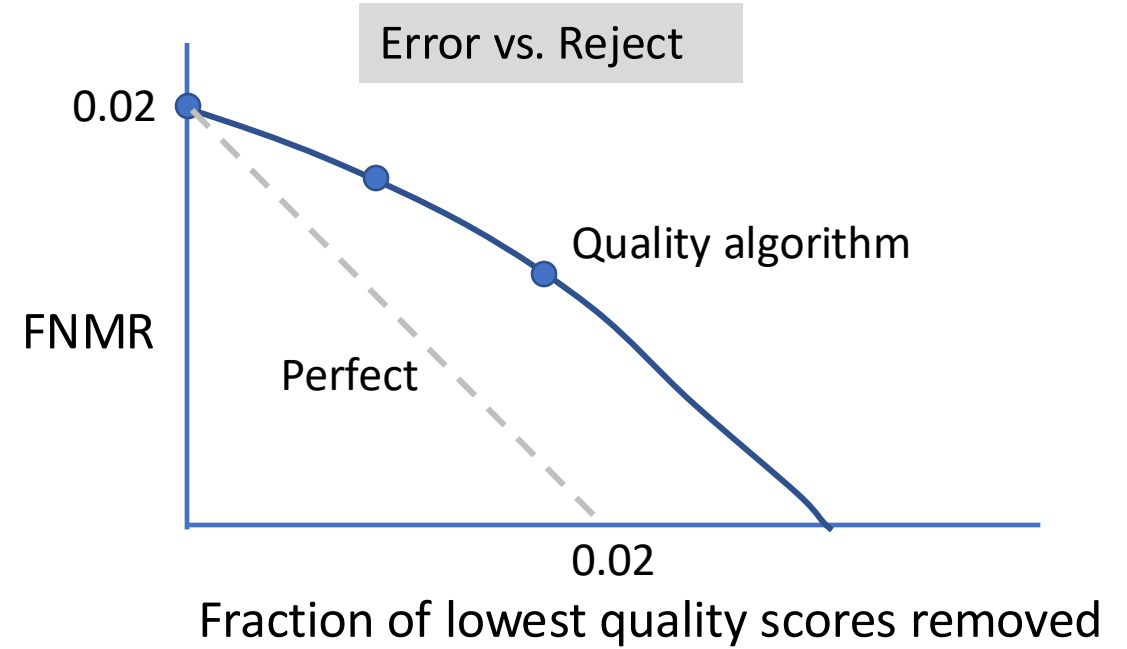
Quality Values as Predictors of Genuine Scores– Error vs. Reject

Quality algorithm

Ref	Probe	Quality
		97
		94
		74
		68
		57
		32
		29
		27

Recognition algorithm

MATCH SCORE	MATCH YES?
0.97	TRUE
0.91	TRUE
0.60	FALSE
0.85	TRUE
0.81	TRUE
0.72	FALSE
0.90	TRUE
0.57	FALSE



FNMR is ideally reduced as quality algorithm is used to discard low quality probes.